

Leonardo Pozzati

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Profile

Senior Software Engineer and Systems Engineer with over 10 years of experience specializing in low-latency, performance-critical systems across fintech and high-tech industries. Proven track record of building proprietary FX trading infrastructure processing 0.8M+ messages/sec with sub-250 μ s latency.

Combines a rigorous academic foundation in Mathematical Physics with deep expertise in C++17/20, lock-free concurrency, and Python data analysis pipelines.

Education

Università degli Studi Roma Tre

Roma, Italy

Master's Degree in Mathematical Physics – 110/110 cum Laude

- Research on the relaxation to the equilibrium of supersaturated gases through critical droplets in metastable thermodynamic systems.
- Designed and implemented stochastic and deterministic mathematical models; developed high-performance C++ simulations and executed massive data analysis pipelines in Python within Non-Equilibrium Statistical Mechanics.

Università degli Studi Roma Tre

Roma, Italy

Bachelor's Degree in Physics

Technical Skills

Programming Languages: C, C++, Python, Bash, SQL, Linux, OOP and UML.

Quantitative & Analytics: Mathematical Physics, Statistical Mechanics, Time-Series Analysis, Machine Learning (MLP, LSTM, TCN, Transformers), Signal Evaluation.

Low-Latency & Systems: Multi-threading, Lock-free Concurrency, Memory Management (NUMA-aware, custom cache-aligned allocators), Linux Kernel Tuning, Linux epoll event loops.

Tools & Infrastructure: ZeroMQ, CI/CD (Jenkins), Git, Docker, Profiling (perf, Flamegraphs, heaptrack), Agile Methodologies.

Professional Experience

Thermo Fisher Scientific

Eindhoven, Netherlands

Software Designer

2025 – Present

- Develop and maintain core automation software for Transmission Electron Microscopes (TEMs) applied to semiconductor chip anomaly detection.
- Collaborate on low-level architectural design, development, and testing of efficiency-critical software modules utilizing C/C++, Python, and Bash in Agile environments.

ASML

Veldhoven, Netherlands

Senior Software Engineer - Universal Light System

2024

- Joined a core team responsible for building a new universal light system for ASML's lithography machines.
- Designed and implemented software modules using C/C++, with scripting in Python and Bash.
- Followed Agile practices to coordinate development, testing, and integration from the ground up.

ASML

Veldhoven, Netherlands

Senior Software Engineer - Safety and Diagnostic

2022 - 2024

- Assigned to the core Safety and Diagnostic team for the ASML's High-NA EUV lithography platform.
- Improved system robustness by integrating, refactoring, and testing both new and legacy code.
- Collaborated with the T&I (Test & Integration) team to integrate CI/CD pipelines and streamline deployments in the way of working of the team.
- Authored the software design documentation for safety-critical components.
- Collaborated with mechanical engineers in the testing and validation of multiple physical machine subsystems, providing software support, diagnostics and issue investigation.

ProMarket

Senior Software Engineer / Quant Developer

Rome, Italy
2018 – 2021

- o Built and owned proprietary high-frequency FX trading infrastructure supporting internal prop trading teams deploying firm capital.
- o Designed and implemented a low-latency routing and messaging framework using C++17, Python, and ZeroMQ (pub/sub, dealer/router, binary star), achieving consistent production latencies of 150–250 μ s.
- o Reduced end-to-end median messaging latency by 50% (from 400 μ s to 200 μ s) via CPU pinning, NUMA-aware workload partitioning, `boost::lockfree::queue`, and lean serialization formats.
- o Scaled messaging throughput from 400K to over 0.8M messages/sec by implementing epoll-based event loops (EPOLLONESHOT), custom cache-aligned allocators, and continuous profiling using `perf`, Flamegraphs, and `heaptrack`.
- o Refactored the real-time risk-checking module with a lock-free memory pool, cutting heap allocation volume under peak load by 30% and stabilizing memory footprint.
- o Resolved market-open production latency spikes by replacing inefficient `gettimeofday()` calls with `clock_gettime(CLOCK_MONOTONIC_RAW)`, successfully reducing latency jitter by 40%.
- o Developed orchestration and coordination services for internal strategy deployment, state management, and system diagnostics using Python, Bash, and Docker.
- o Contributed to a proprietary financial sentiment analysis engine, extracting quantitative trading signals from market data and news feeds to evaluate pre-production signal quality.
- o Collaborated directly with traders during initial production rollouts, aligning system performance and operational requirements with trading desk needs.
- o Supported live production for 18+ months, managing 10–15K orders/day across 3–5 FX trading venues with multiple parallel execution strategies.

Key Projects

WOPR | Personal Quant & ML Trading Project

- o Engineered an end-to-end quantitative pipeline in Python for processing CME sessionized Nasdaq futures data.
- o Implemented range-bar feature engineering, Time-to-Horizon (TTH) label generation, and trained multi-model deep learning architectures including MLP, LSTM, TCN, and Transformers.
- o Enhanced pipeline reproducibility and system robustness by unifying core labeling logic and integrating checkpoint-driven scaling and metadata tracking across train, evaluation, and inference cycles.

Domain-Specific Chatbot Training | Educational Software Engineering Notebook

- o Developing a structured, reference-heavy knowledge base and data compilation pipeline focused on optimizing LLMs for specific engineering domains.

Languages

Italian: Native | **English:** C1 - Academic IELTS | **Dutch:** Self-studying